

Elaina Mann

(972) 980 - 9936 • elainaymann@gmail.com • [elainamann.me](mailto:elainaymann@gmail.com)

EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Engineering in Computer Engineering

May 2026

Bachelor of Science in Engineering in Robotics

May 2026

Graduated *cum laude*

Relevant coursework: Advanced Embedded Systems, Intro Embedded Systems Design, Computer Organization, Logic Design, Electrical Circuits, Sensors & Signals, Hardware/Coding for IoT

TECHNICAL SKILLS

Programming	C, C++, Python, Verilog, MATLAB, Julia
Embedded & Platforms	STM32, ESP32, Arduino, Raspberry Pi, ROS2, FreeRTOS, UART, I ² C, SPI, CAN, Linux, Git
Hardware & Design	Altium, KiCad, PCB Design, CAD, Lab Instrumentation
Core Competencies	Embedded Systems, Electronics, Hardware Integration, Software Design, Rapid Prototyping

WORK EXPERIENCE

University of Michigan Department of Performing Arts Technology

Ann Arbor, MI

Mechatronics Team Research Assistant

Jan 2024 – Dec 2024

- Restructured robot electrical system by adding ADCs, H-bridges, fuses, and wire management, leading to optimized power usage and increased safety.
- Designed embedded C++ algorithms using RTOS to map musician movement to robot movement for intuitive robot response.
- Created a 3D CAD model of the robot, enabling a more streamlined mapping of musician movement to robot movement and opening the door to simulation opportunities.

Panasonic

Newark, NJ

Automation Engineering Intern

Jan 2024 – Oct 2024

- Built an application that successfully centralized document requests and allowed real-time chat communication between 200+ members of North American Panasonic departments.
- Designed the internal web application based on user feedback and key stakeholder meetings to understand requirements.
- Learned Power Fx independently to develop complex application logic, database interactions, automation, and security features.

PROJECT EXPERIENCE

University of Michigan Robotic Submarine Team

Ann Arbor, MI

Software/Electrical Team Member

Aug 2022 – May 2026

- Led hardware and embedded software development for a hydrophone localization system, integrating custom electronics and real-time sensing to achieve a top-5 placement at the international RoboSub competition.
- Designed custom PCBs in Altium and mechanical structures in CAD, improving electrical system organization, component integration, and maintainability.
- Developed a Python navigation state machine for autonomous submarine control, implementing sensor-driven behaviors.

President

May 2024 – May 2025

- Led a 30+ member robotic submarine team, aligning electrical, mechanical, and administrative subteams.
- Directed weekly meetings, creating agendas and presentations to track progress and resolve blockers.
- Secured \$20k+ in sponsorships, managed budgeting, and coordinated travel logistics for 20 members.

Wireless Blindspot Detection System for Semi-Trucks

Ann Arbor, MI

Advanced Embedded System Term Project | Sponsored by Infineon

Aug 2025 – Dec 2025

- Designed 2 PCBs featuring ESP32 microcontrollers, ultrasonic sensors, boost converters, and LiPo battery management circuits.
- Implemented Bluetooth architecture with FreeRTOS, enabling Barnacle sensor units to report vehicle detection status.
- Developed iOS companion app supporting device configuration, driving mode notifications, and live diagnostic logging.

Mini Ballast Submarine

Ann Arbor, MI

Embedded System Term Project

Aug 2025 – Dec 2025

- Integrated IMU, pressure, temperature, and flow sensors using UART and I²C communication protocols with STM32-based embedded system for real-time monitoring.
- Developed CAN bus communication between submarine and ground station, enabling bidirectional sensor data and control.
- Implemented actuator control using H-bridges and PWM for buoyancy adjustment and dual-thruster propulsion.